

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		award (1) for any of following - all have 19 protons but 20, 21 and 22 neutrons - all have 19 protons but different numbers of neutrons - all have same number of protons but 20, 21 and 22 neutrons all have same number of protons but different numbers of neutrons - neutral answer ignore references to electrons		1		1		
		(ii)		39.1 (3) 39.13468 (2) award (1) for correct substitution (39 × 93.1) + (40 × 0.0122) + (41 × 6.88)	1	2		3	3	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		award (1) for any two similarities • both float • both move • both bubble on surface • both produce hydrogen / gas • both form hydroxides / alkaline solutions award (1) for any two differences • potassium melts into ball (but lithium doesn't) • potassium ignites / burns (but lithium doesn't) • potassium bubbles / moves more rapidly (than lithium) • potassium is more reactive (than lithium)	2			2		2
		(ii)		$2K + 2H_2O \rightarrow 2KOH + H_2$ reactants (1) products (1) balancing (1) - reactants and products must be correct for balancing mark to be awarded		3		3		
				Question 7 total	3	6	0	9	3	2